Trainable/fitted
Frozen
Labeled anomaly
Labeled normal
Unlabeled target/query
Queried node
Normal prototype
Abnormal support prototype
High anomaly score
Low anomaly score
Element-wise product
Residual addition
Concatenation
Information flow
Source-domain Training
Source(labeled)
Target(unlabeled)
17-D
772-D
t
Invariant feature construction
16-D Attribute statistics
8-D Stuctural-temporal
24-D
Target 24-D
Source 24-D
Dynamic GNN
$G_t=(\mathbf{X}_t,\mathbf{A}_t)$
$\mathbf{x}_t$
$\mathbf{m}_{t-1}$
$\mathbf{W}_m$
GNN(G_t)
$\mathbf{h}_t^{\mathrm{sp}}$
$G_{t-1}\rightarrow G_t$
Evolution-aware update
$\mathbf{m}_t$
P_{sp}:Linear
$\mathbf{g}_t\in\mathbb{R}^{32}$
Causal TCN
$\mathbf{x}_{1:T}\in\mathbb{R}^{T\times 24}$
past→present(T)
TCN block,d=1
TCN block,d=2
TCN block,d=4
$\mathbf{h}_{1:T}^{\mathrm{temp}}$
P_{temp}:Linear
$\mathbf{s}_{1:T}\in\mathbb{R}^{T\times 32}$
Residual Interaction
Projected temporal states
$\mathbf{s}_{t-1}\in\mathbb{R}^{32}$
$\mathbf{s}_t\in\mathbb{R}^{32}$
$\Delta_{\mathrm{temp}}=\mathbf{s}_t-\mathbf{s}_{t-1}$
$\Delta_{\mathrm{temp}}\in\mathbb{R}^{32}$
Linear $\mathbf{W}_{\mathrm{temp}}$
$\mathbf{u}_t\in\mathbb{R}^{32}$
Concat(||)
Linear $\mathbf{W}_f$
$\mathbf{r}_t\in\mathbb{R}^{32},\quad t=2,\dots,T$
$\mathbf{r}_{1:T}$
$\mathbf{r}_1=\mathbf{0}\in\mathbb{R}^{32}$
Projected spatial states
$\mathbf{g}_{t-1}\in\mathbb{R}^{32}$
$\mathbf{g}_t\in\mathbb{R}^{32}$
$\Delta_{\mathrm{sp}}=\mathbf{g}_t-\mathbf{g}_{t-1}$
$\Delta_{\mathrm{sp}}\in\mathbb{R}^{32}$
Linear $\mathbf{W}_{\mathrm{st}}$
$\mathbf{v}_t\in\mathbb{R}^{32}$
Hierarchical Fusion
$\mathbf{r}_1=\mathbf{0}$
$\mathbf{r}_{1:T}\in\mathbb{R}^{T\times 32}$ (from Residual)
$\mathbf{h}_{1:T}^{\mathrm{sp}}\in\mathbb{R}^{T\times 32}$ (GNN skip)
$\mathbf{x}_{1:T}\in\mathbb{R}^{T\times 24}$ (input skip)
1 … t … T
Concat (||)
$\mathbf{c}_t=[\mathbf{r}_t\|\mathbf{h}_t^{\mathrm{sp}}\|\mathbf{x}_t]\in\mathbb{R}^{88}$
Fusion MLP
Linear 88→64
ReLU + Dropout
Linear 64→32
Linear 32→32
$\mathbf{z}_t\in\mathbb{R}^{32}$
$\mathbf{z}_1$ $\mathbf{z}_2$ $\mathbf{z}_T$
$t=1,\dots,T$ (same MLP for all t)
mean over t
$\overline{\mathbf{z}}\in\mathbb{R}^{32}$
$\overline{\mathbf{z}}=\frac{1}{T}\sum_{t=1}^{T}\mathbf{z}_t$
Source-domain Training Objectives
$\mathcal{L}_{det}$ Supervised detection
$\overline{\mathbf{z}}_i\in\mathbb{R}^{32}$
$\mathbf{p}_N\in\mathbb{R}^{32}$ (trainable)
$d_i^p=1-\cos(\overline{\mathbf{z}}_i,\mathbf{p}_N)$
$a_i=\alpha w_p(d_i^p-\tau_p)+\alpha w_t(\tilde{v}_i-\tau_t)$
a_i (logit)
BCEWithLogits
$\mathcal{L}_{\mathrm{det}}$
$y_i\in\{0,1\}$ (source label)
$\mathbf{z}_{1:T,i}$
$\hat{\mathbf{z}}_{t,i}=\frac{\mathbf{z}_{t,i}}{\|\mathbf{z}_{t,i}\|_2}$
$v_i=mean\,\|\hat{\mathbf{z}}_{t,i}-\hat{\mathbf{z}}_{t-1,i}\|_2$
$\tilde{v}_i$
$\mathcal{L}_{proto}$ Normal-prototype compactness
$y_i=0$ only
$\overline{\mathbf{z}}_1$ $\overline{\mathbf{z}}_2$ $\overline{\mathbf{z}}_n$
$\mathbf{p}_N$ (trainable) $\mathbf{p}_N\in\mathbb{R}^{32}$
$\mathcal{L}_{\mathrm{proto}}=\underset{y_i=0}{mean}\,[1-\cos(\overline{\mathbf{z}}_i,\mathbf{p}_N)]$
$\mathcal{L}_{self}$ Temporal consistency
$\hat{\mathbf{z}}_{1:T,i}$ (L2 normalized fused sequence)
normal volatility μ_N
abnormal volatility μ_A
m
t
$\mathcal{L}_{\mathrm{self}}=\mu_N+ReLU(m+\mu_N-\mu_A)$
source labels only
$\mathcal{L}=\mathcal{L}_{\mathrm{det}}+\lambda_{\mathrm{proto}}\mathcal{L}_{\mathrm{proto}}+\lambda_{\mathrm{self}}\mathcal{L}_{\mathrm{self}}$
Zero-shot Inference
Target 24-D sequence
G_1 G_2 … G_T
$\mathbf{x}_{1:T}\in\mathbb{R}^{T\times 24}$
Frozen SPAD backbone
Dynamic GNN
Causal TCN
Input d=1 d=2 d=4 past t
Residual Interaction
Hierarchical Fusion
$\mathbf{z}_{1:T}$
L2 normalize
$\hat{\mathbf{z}}_{t,i}=\frac{\mathbf{z}_{t,i}}{\|\mathbf{z}_{t,i}\|_2}$
$\hat{\mathbf{z}}_{1:T}$
Local disagreement
$\ell_{t,i}=1-\cos(\hat{\mathbf{z}}_{t,i},\boldsymbol{\mu}_{t,i}^{\mathcal{N}})$
Recency weighting
$t=1$ $w_1=1$ $\ell_{1,i}$
$t=2$ $w_2=2$ $\ell_{2,i}$
$t=T$ $w_T=T$ $\ell_{T,i}$
$\sum$ q_i
Recent snapshots weigh more
Percentile rank
sort
$\rho_i=\frac{rank(q_i)}{\max(N-1,1)}$
Anomaly score
$a_i=\alpha(\rho_i-0.5)$
$s_i=\sigma(a_i)$
Few-shot Adaptation & Inference
Target 24-D sequence
Frozen SPAD backbone
Labeled target support S
G_T
y = 0
y = 1
$S\cap Q=\varnothing$
Frozen evidence bank
Zero-shot anchor
Local disagreement
Recency weighting
Percentile rank
Percentile-ranked cues
recency local
uniform local
latest local
transport residual
prototype distance
mean fused
latest fused
temporal change
Support-only CV selector
Keep zero-shot $\psi_i=0$
Logistic score readout
Support prototype margin
$\mathbf{p}_N^S$ $\mathbf{p}_A^S$ ψ_i
Use correction only if $\Delta_{\mathrm{CV}}>\delta$
Conservative fusion
$s_i^{\mathrm{FS}}=(1-\alpha_{\mathrm{FS}})\rho_i^{\mathrm{rec}}+\alpha_{\mathrm{FS}}\psi_i$
Zero-shot anchor ρ_i^{rec}
Support correction ψ_i
$\alpha_{\mathrm{FS}}=\begin{cases}\alpha^*, & \Delta_{\mathrm{CV}}>\delta,\\ 0, & \Delta_{\mathrm{CV}}\le\delta.\end{cases}$
Few-shot anomaly score
Query Q